Name:	Class:	Date given:
		Date due in:

Light

1. Complete the diagrams below by drawing arrows to show how the light wave behaves at each object.

   a. a glass block b. a glass block c. a mirror d. a prism



2. Use the words below to fill in the gaps.[8]

flame	luminous	moon	non-luminous	reflect	sun	transmit	window

_____ objects emit their own light source. Examples include the _____ or a _____. _____ objects do not emit their own light; however, they may _____ light from their surface or _____ light through. Examples include the _____ and a _____.

3. Below are the colours of the visible spectrum. Write them into the correct order of shortest to longest wavelength.[5]

   red violet indigo green yellow orange blue

   _____

4. State the colour that each object would appear: [5]

   a. A black object under white light _____

   b. A white object under red light _____

   c. A green object under white light _____

   d. A red object under blue light _____

   e. A red object under yellow light _____

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5. Peter shines his torch on a mirror at an angle of 36°. What angle will the light ray be reflected at? [1]

6. Peter shines his torch on a mirror where the glass has been shattered. Describe how the light is reflected. [1]

7. What is specular reflection? [1]

8. In which medium will light travel the fastest? Circle one [1]

   solid liquid gas vacuum

Learning Outcomes (tick if achieved)

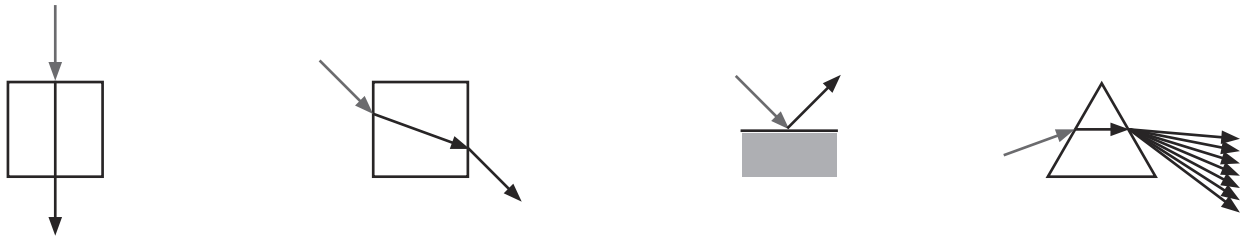
Q1	I can represent reflection and refraction in diagrams	
Q2	I can describe objects as luminous or non-luminous	
Q3	I can recall the colours of the visible spectrum	
Q4	I can describe how colours are perceived	
Q5	I know the law of reflection	
Q6,7	I can describe types of reflection	
Q8	I understand how light energy travels	

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Light **Answers**

1. Complete the diagrams below by drawing arrows to show how the light wave behaves at each object.

   a. a glass block b. a glass block c. a mirror d. a prism



2. **Luminous** objects emit their own light source. Examples include the **sun** or a **flame**. **Non-luminous** objects do not emit their own light; however, they may **reflect** light from their surface or **transmit** light through. Examples include the **moon** and a **window**.

3. Below are the colours of the visible spectrum. Write them into the correct order of shortest to longest wavelength.[5]

   red orange yellow green blue indigo violet

4. State the colour that each object would appear: [5]

   a. A black object under white light **black**

   b. A white object under red light **red**

   c. A green object under white light **green**

   d. A red object under blue light **black**

   e. A red object under yellow light **black**

5. Peter shines his torch on a mirror at an angle of 36°. What angle will the light ray be reflected at? [1]

   36°

6. Peter shines his torch on a mirror where the glass has been shattered. Describe how the light is reflected. [1]

   The light appears dull, light is scattered

7. What is specular reflection? [1]

   This is when all the light reflects off a surface at the same angle.

8. In which medium will light travel the fastest? Circle one [1]

   solid liquid gas (vacuum)